



U.S. Department  
of Transportation

Research and  
Special Programs  
Administration

3AL CYLINDER

400 Seventh Street, S.W.  
Washington, D.C. 20590

DOCKETS UNIT  
(202)366-5046 BUS.  
(202)366-3753 FAX

CYLINDERS 3-AL  
SEE LIST BELOW

D.O.T.E- 6498

7042

8107

8364

8422

6616

6668 ✕

8404

SEE SECTION 173.23  
49 CFR CODE OF REGULATIONS

§ 173.23 Previously authorized packaging.

(a) When the regulations specify a packaging with a specification marking prefix of "DOT," a packaging marked prior to January 1, 1970, with the prefix of "ICC" may be used in its place if the packaging otherwise conforms to applicable specification requirements.

(b) [Reserved]

(c) After July 2, 1982, a seamless aluminum cylinder manufactured in conformance with and for use under DOT exemption E 6498, E 7042, E 8107, E 8364, or E 8422, may be continued in use if marked before or at the time of the next retest with the specification identification "3AL" immediately above the exemption number, or the DOT

mark (e.g., DOT 3AL 1800) is added in proximity to the exemption marking.

(d) Cylinders (spheres) manufactured and marked DOT-E 6616 prior to January 1, 1983, may be continued in use if marked before or at the time of the next retest with the specification identification "4BA" near the exemption marking.

(e) After October 1, 1984, cylinders manufactured for use under exemptions ✕ ~~DOT E-6668 or E-8404~~ may be continued in use, and must be marked "DOT-4LXXXXYY" (XXX to be replaced by the service pressure, YY to be replaced by the letters "AL", if applicable) in compliance with Specification 4L (§178.57 of this subchapter) on or before January 1, 1986. The "DOT-4LXXXXYY" must appear in proximity to other required specification markings.

(f) An MC 331 cargo tank motor vehicle must conform to structural integrity requirements in §178.337-3 or to corresponding requirements in effect at the time of manufacture.

(g) A non-bulk packaging manufactured, tested, marked, and certified on or before September 30, 1996, in accordance with the applicable provisions of subparts L and M of part 178 of this subchapter in effect on September 30, 1995, may be used as authorized by this subchapter if the packaging conforms to all requirements applicable at the time of manufacture. In addition, such a packaging may be reused as authorized by §173.28 without a nominal thickness marking, if it conforms to the minimum thickness criteria prescribed in §173.28(b)(4).

[Amdt. 173-3, 33 FR 14921, Oct. 4, 1968, as amended by Amdt. 173-94, 41 FR 16063, Apr. 15, 1976; Amdt. 173-152, 47 FR 26633, June 21, 1982; Amdt. 173-16, 48 FR 50460, Nov. 1, 1983; Amdt. 173-176, 49 FR 24689, June 14, 1984; Amdt. 173-180, 49 FR 42735, Oct. 24, 1984; Amdt. 173-224, 55 FR 52610, Dec. 21, 1990; 57 FR 45460, Oct. 1, 1992; Amdt. 173-240, 60 FR 17400, Apr. 5, 1995; Amdt. 173-242, 60 FR 26805, May 18, 1995]



DEPARTMENT OF TRANSPORTATION  
HAZARDOUS MATERIALS REGULATIONS BOARD  
WASHINGTON, D.C. 20590

SPECIAL PERMIT NO. 6668

This special permit is issued pursuant to 49 CFR 170.15 of the Department of Transportation (DOT) Hazardous Materials Regulations, as amended, to authorize shipments of a liquefied, compressed gas under conditions as prescribed herein. This permit does not relieve any shipper or carrier from compliance with any requirement of the DOT Regulations, except as specifically provided for herein.

Standard special permit requirements and conditions relating to package markings, preparation of shipping papers, shipping experience reports, etc., are published in 49 CFR 171.6. These requirements are part of this special permit.

1. BASIS. Petition dated April 4, 1972 and supplemented by letters dated May 22 and July 17, 1972 by Union Carbide Corporation, Tonawanda, New York.
2. COMMODITY. Oxygen.
3. PROPER SHIPPING NAME (49 CFR 172.5). Oxygen, pressurized liquid.
4. REGULATION AFFECTED. 49 CFR 173.304(a)(2), 178.57.
5. AUTHORIZED SHIPPER. The petitioner identified above, and any other shipper who registers his identity with and receives acknowledgement from this Board and has a copy of the special permit.
6. PACKAGING PRESCRIBED. An insulated, welded cylinder complying with DOT Specification 4L (49 CFR 178.57) in all respects except as follows:
  - a. Instead of the insulating material being covered with a steel jacket, it is covered with a jacket made from Type 3003 aluminum, 0.070 inch thick.
  - b. The insulating material must be laminar type insulation consisting of alternate layers of aluminum foil and micro glass-fiber mat continuously wound over the stainless steel cylinder.
  - c. The safety relief capacity must be sized on a bare tank basis.
  - d. Cylinder must not be marked "DOT-4L \*\*\*" per 178.57-20(a)(1) but, instead, must be marked "DOT SP 6668". All other marking requirements of 178.57-20 apply.
7. SPECIAL PACKAGING REQUIREMENT. Cylinder is limited to a maximum water weight capacity of 100 pounds.